

Amira Avizo Protocol for Converting CNN Distance Maps to Cell-Instance Segmentation

Three orthogonal CNN distance maps (**5.5p-ROI2xy.filtered**, **5.5p-ROI2xz.filtered**, **5.5p-ROI2yz.filtered**) are denoised, fused, and split into individual cell instances using a marker-controlled watershed. Only complete cells are retained.

1. Apply a Amira's **Gaussian Filter** function to each map, smoothing only along the axis orthogonal to its prediction plane:
 - a. **xy** map: smooth in **z**
 - b. **xz** map: smooth in **y**
 - c. **yz** map: smooth in **x**
2. Combine the three distance maps by summing voxel values using Amira's Arithmetic module.
3. Use Amira's **Resample** modules to down sample by a factor of 2. This is not completely necessary but it will reduce computational time by a factor of 2^3 , and empirically seems to have little effect on segmentation performance.
4. Apply Amira's **H-Extrema Watershed** module using Basins as the seed.
5. Use Amira's **Remove Small Spots** module to remove noise.
6. Use Amira's **Border Kill** module to remove cells not fully included in the volume

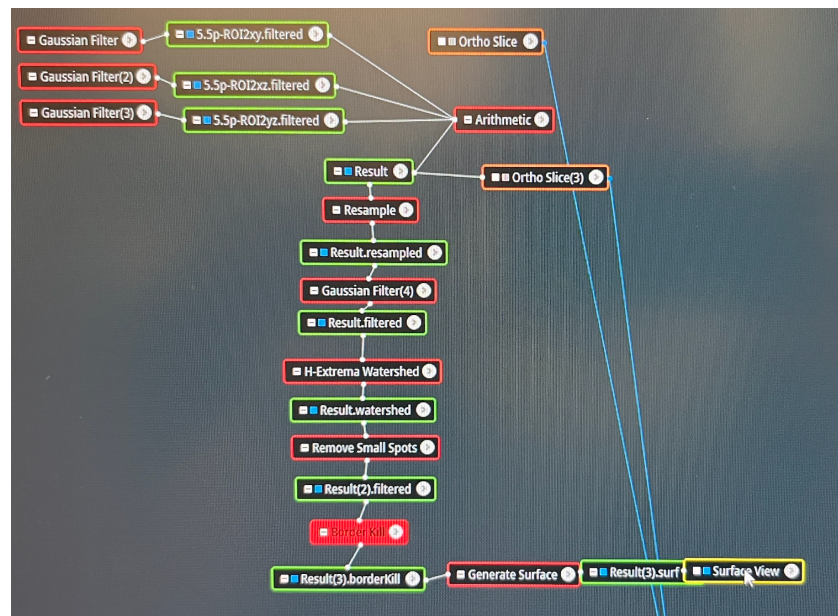


Figure 1: Amira pipeline for instance level cell segmentation